

Preparation note (remove before submission): This draft consolidates the experimental results supplied by the authors into the IEEE structure of the original manuscript. Entries marked **[VERIFY]** indicate a numerical or provenance inconsistency identified during the audit (see the accompanying audit report) that could not be resolved without access to the original per-sample logs. Figure 1 is intentionally left as a placeholder for manual insertion, as requested.

Severity-Aware and Uncertainty-Guided Deep Learning Framework for Automated Solar Panel Defect Inspection

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Abstract

Automated inspection of photovoltaic (PV) modules using electroluminescence (EL) imaging and deep learning has attracted substantial research interest, yet most published pipelines assume that a single, well-composed cell image is already available, treat defect severity as an unordered category, and return a point prediction with no signal of its own reliability. This paper reports an integrated, severity-aware and uncertainty-guided inspection framework combining YOLO-based panel/cell localization, quality-gated multi-frame candidate selection, an ordinal severity classifier, Monte Carlo (MC) dropout uncertainty estimation, and a maintenance-priority decision layer, evaluated on the public ELPV electroluminescence benchmark (2,624 grayscale cell images from 44 modules). Among five candidate backbones, EfficientNet-B0 trained with a rank-consistent ordinal loss obtained the strongest four-class severity accuracy (95.6%, macro-F1 93.5%, severity MAE 0.164, quadratic weighted kappa 0.934), improving MAE by about 23% relative to an otherwise identical cross-entropy classifier (0.213). An incremental ablation indicated that quality-gated candidate-frame selection, ordinal severity learning, and MC-dropout-based calibration each contributed measurable, non-redundant gains, jointly lowering expected calibration error from 8.9% to 2.9% and raising error-detection AUROC from 0.71 to 0.88. A selective-prediction analysis showed that withholding the 20% most uncertain predictions raised retained-set accuracy to 99.0%, an accuracy-coverage trade-off rather than a change in overall model accuracy. A YOLO-based localizer, trained on a separate panel/cell bounding-box dataset, reached 96.1% mAP@0.50. Under simulated blur, noise, brightness/contrast, and JPEG degradation, accuracy losses grew with corruption severity, most sharply under blur, and the full framework degraded less than a single-frame, softmax-only baseline at every tested severity. A metric consistency audit performed on the reported tables identified a discrepancy between the supplied confusion matrix and the corresponding class-wise precision/recall/F1 table; this is flagged explicitly rather than silently corrected. These results were obtained on one comparatively small, laboratory-curated dataset using synthetic multi-frame and degradation protocols rather than field-acquired sequences, and the reported gains should be read within that scope.

Index Terms— Photovoltaic inspection, electroluminescence imaging, defect severity classification, ordinal regression, uncertainty estimation, Monte Carlo dropout, image quality assessment, selective prediction, predictive maintenance, deep learning.

I. Introduction

A. Background and Motivation

Photovoltaic (PV) capacity continues to expand as part of the broader shift toward renewable energy, and the reliability of individual modules is a direct driver of long-term energy yield. PV cells accumulate defects during manufacturing and field operation, including microcracks, finger interruptions, dislocations, and cell-to-cell disconnections, several of which are invisible under ordinary white-light inspection [1], [3].

Electroluminescence (EL) imaging, which captures the infrared emission of a forward-biased module, is widely regarded as the most informative non-destructive modality for revealing such internal defects [1], [3], but manual interpretation by trained personnel remains the practical standard in many facilities, and it is slow, costly at plant scale, and subject to inter-observer variability [28].

B. Problem Statement

Most reported EL classifiers are trained and evaluated on curated datasets in which each cell is already represented by a single, well-framed image. In field deployment that image is itself the product of an acquisition process; a classifier that ignores this step can still be handed a blurred, poorly exposed, or occluded frame and asked to produce a defect label regardless. Most systems also output a single label with no accompanying confidence signal, which limits their use for prioritizing maintenance across a large module fleet.

C. Limitations of Existing Approaches

On the ELPV benchmark, Tella et al. [26] report four-class ensemble accuracies of 68.36–68.31% and 73.02% binary accuracy with ResNet-18; Sudervel et al. [27] report 96.4% four-class accuracy with a hybrid EfficientNet-B0+SVM pipeline; Demir [28] reports the strongest published figures to date, 98.17% binary and 97.02% four-class, using a residual-attention CNN combined with a ReliefF-weighted subspace SVM stage. Detection-oriented work such as YOLO-PV [5] and transformer-based classifiers such as PVEL-ViT [9] extend accuracy and localization further, and field-oriented studies [4] confirm that EL-based classification remains sensitive to real operating conditions. None of these studies, nor the broader review in [3], combines an explicit acquisition-quality stage, an ordinal severity target, and a calibrated uncertainty estimate alongside its predictions.

D. Research Gap

This paper does not claim novelty for any individual technique reviewed in Section II; localization, EL classification, ordinal regression, and uncertainty estimation are each separately well established. The gap addressed here is their combined use within a single evaluable pipeline that also feeds a maintenance-priority decision, tested together under a controlled acquisition- and degradation-aware protocol rather than assumed to transfer unchanged from clean-image benchmarks.

E. Research Questions

RQ1. Can automated localization and quality-aware acquisition improve downstream classification reliability?

RQ2. Does an ordinal severity formulation outperform conventional categorical classification? **RQ3.** Can uncertainty estimation identify predictions that should be referred for human review? **RQ4.** How robust are candidate backbones under realistic degradation (blur, noise, brightness/contrast, compression)? **RQ5.** What is the accuracy–computational-cost trade-off across candidate backbones?

Each question is answered explicitly in Sections VII and VIII against the completed experiments; none is treated as fully resolved beyond what the reported tables support.

F. Contributions

(i) an integrated inspection pipeline combining panel/cell localization, quality-aware multi-frame candidate selection, severity-aware ordinal classification, MC-dropout uncertainty estimation, selective prediction, and a maintenance-priority decision layer; (ii) an experimental comparison of ordinal versus categorical severity learning on the ELPV probability labels; (iii) a quality-gated candidate-frame selection strategy evaluated under a simulated multi-frame protocol; (iv) a systematic robustness evaluation under simulated blur, noise, brightness/contrast, and compression; (v) an uncertainty and selective-prediction analysis reporting calibration and accuracy-coverage trade-offs; and (vi) an accuracy-efficiency comparison across five CNN backbones. No individual component is presented as newly invented; the contribution is their joint evaluation.

[FIGURE 1 ABOUT HERE: OVERALL ARCHITECTURE OF THE PROPOSED SEVERITY-AWARE AND UNCERTAINTY-GUIDED SOLAR PANEL INSPECTION FRAMEWORK]

Figure 1. Overall architecture of the proposed framework: image acquisition → YOLO localization → candidate frame generation/acquisition → image quality assessment → quality gating → best-frame selection → preprocessing → EfficientNet-B0/backbone classification → ordinal severity prediction → MC-dropout uncertainty → selective prediction → maintenance priority. *(Diagram to be inserted manually by the authors; not generated here.)*

II. Related Work

A. Conventional PV Inspection Modalities

IR thermography, EL imaging, and I-V curve analysis are the principal defect-detection modalities used in PV inspection [28]. I-V analysis characterizes module-level performance but cannot localize individual defective cells; IR thermography visualizes thermal anomalies such as hot spots but has limited resolution for microcracks [28]; EL imaging offers the highest spatial resolution for cell-level internal defects and is the modality used in this study [1], [3].

B. Deep-Learning-Based EL Classification

Deitsch et al. [1] established the ELPV benchmark with CNN/SVM baselines over probability-based labels. Tella et al. [26] fine-tuned eight CNN/transformer backbones (AlexNet, SENet, GoogleNet, Xception, ViT, Darknet53, ResNet-18, SqueezeNet) and combined them with voting and bagging ensembles. Sudervel et al. [27] paired six CNN/transformer feature extractors (AlexNet, VGG16, ResNet-18/50, EfficientNet-B0, Swin Transformer, ConvNeXt) with classical classifiers (SVM, logistic regression, KNN, XGBoost). Demir [28] combined a residual-attention CNN trained from scratch with a ReliefF-based subspace-weighted SVM (RSWS) stage, reporting the highest published ELPV accuracies to date and validating the approach on an independent EL dataset. These three studies are the closest points of comparison for the classification component evaluated here.

C. Detection, Attention, and Transformer Approaches

YOLO-based detectors localize panels or cells rather than classify severity: YOLO-PV [5] reported 94.55% AP on the ELPV benchmark, compared with 78% AP for an earlier YOLO variant reported in the same study; subsequent variants such as YOLOv8-GD [6] and C2DEM-YOLO [7] targeted background complexity and scale variation, and [8] proposed an efficient anchor-free detector. Attention and transformer classifiers include Demir's residual-attention CNN [28], the token-selective vision transformer PVEL-ViT [9], which reports 95.84% accuracy on an EL dataset (2.1–2.4 percentage points above EfficientNetV2-m and EfficientViT-b2 baselines),

and the deformable transformer PDeT for defect segmentation [10]; Sudervel et al. [27] found transformer backbones competitive with, but not uniformly superior to, CNN-based hybrids on ELPV.

D. Severity, Robustness, and Uncertainty

ELPV already encodes an ordinal condition score (0, 0.33, 0.67, 1) [1], [2], [26], yet every classification study reviewed above trains it with unordered categorical cross-entropy. Rank-consistent ordinal regression (CORAL [15], CORN [16]) exploits such ordering elsewhere but has not, to our knowledge, previously been applied to PV severity labels. Separately, Hendrycks and Dietterich [14] showed that clean-image accuracy is a poor predictor of robustness to common corruptions, and Lee et al. [17] showed that motion blur alone can collapse a crack-detector's F1-score from 89.4% to 4.5%, with no-reference image-quality metrics able to gate which frames reach the detector. For reliability, MC dropout [11] and deep ensembles [12] are established, scalable uncertainty estimators, and Guo et al. [13] showed that modern networks are typically overconfident but can be recalibrated with temperature scaling. None of these general techniques appears, in combination, in the PV defect-classification literature surveyed in II-B–II-C.

E. Predictive Maintenance Context

Recent reviews of AI-based predictive maintenance for PV assets [18]–[20] emphasize condition monitoring and failure prioritization at the plant level but do not couple cell-level severity classification with a calibrated, uncertainty-aware referral mechanism of the kind evaluated here; this study's maintenance-priority layer (Section IV-K) is positioned as a cell-level complement to that broader literature rather than a replacement for it.

F. Backbone and Loss Foundations

The candidate classification backbones build on established architectures — AlexNet-style CNNs [21], deep residual networks [22], EfficientNet compound scaling [23], and vision transformers [24] — and the localization stage builds on the YOLO family of single-stage detectors [25]. None of these general-purpose architectures is claimed as a contribution of this paper; they are used as established, interchangeable components.

G. Summary of Research Gap

The literature already contains capable detectors, attention/transformer classifiers, and high-accuracy hybrid pipelines for EL-based PV inspection. The gap addressed here is their combination with quality-aware acquisition, ordinal severity, and calibrated uncertainty, evaluated together under a controlled robustness protocol (Table I).

TABLE I. Comparison of Related PV Defect Inspection Studies

Ref.	Model	Severity	Uncertainty	Acquisition
[1]	CNN, SVM	Nominal	No	Curated
[26]	8-model ensemble	Nominal	No	Drone (curated)
[27]	CNN/ViT + SVM/XGB	Nominal	No	Curated
[28]	RA-CNN + RSWS	Nominal	No	Curated
[9]	PVEL-ViT	Nominal	No	Curated

Ref.	Model	Severity	Uncertainty	Acquisition
Proposed	CNN/ViT + ordinal	Ordinal	MC dropout	Multi-frame, quality-gated (simulated)

III. Dataset and Problem Formulation

A. ELPV Dataset

The classification and severity experiments use ELPV [1], [2]: 2,624 grayscale, 8-bit, 300×300-pixel EL images extracted from 44 monocrystalline and polycrystalline modules, each cell labelled by domain experts with a continuous defect probability $p \in \{0, 1/3, 2/3, 1\}$ and a module-type tag. ELPV is used here because it is, to our knowledge, the only public EL benchmark with graded, probability-based labels suitable for an ordinal reformulation, and because it permits direct, like-for-like comparison against the accuracy figures already reported in [1], [26]–[28]. The dataset images are distributed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 license, with the accompanying loader code under Apache License 2.0; this study's use is academic and non-commercial, consistent with that license.

B. Dataset Source and Provenance

ELPV was not collected by the authors of this study. The benchmark was introduced by Buerhop-Lutz et al. at EU PVSEC 2018 [2], with the associated classification study published by Deitsch et al. in *Solar Energy* [1]. The official code and data repository is maintained by the ZAE Bayern group on GitHub [29]; an archival copy is deposited at the Forschungszentrum Jülich data repository [30] **[VERIFY — the archival record's exact content and version were not independently confirmed in this work and should be checked against the GitHub release actually used before submission]**. This paper draws a firm distinction between the original dataset creators [1], [2], the official repository [29], the archival record [30], and this study's own re-use and relabeling of the data for an ordinal severity task. ELPV is single-modality, laboratory-curated, grayscale EL imagery; it is not drone-acquired RGB imagery, and it does not include YOLO-style bounding-box annotations.

C. Class Distribution

Table II reports the class distribution used in this study, following the four-way probability-to-label grouping adopted from [26]. Percentages are computed directly from the counts (1502/2624 = 57.23%, 123/2624 = 4.69%, 298/2624 = 11.36%, 701/2624 = 26.71%).

TABLE II. ELPV Class Distribution and Ordinal Severity Mapping

Class	Original Defect Probability p	Number of Samples	Percentage	Ordinal Severity Level
Functional	0.00	1,502	57.23%	0
Moderate	0.33	123	4.69%	1
Mild	0.67	298	11.36%	2
Severe	1.00	701	26.71%	3
Total	—	2,624	100.00%	—

The four class names are a research-level reformulation of ELPV's continuous probability label and are not the dataset's original semantic categories; ELPV itself provides only the numeric p value and module type [1], [2]. **[VERIFY — naming consistency]**: the ordinal levels 0–3 increase monotonically with p , which is the property the ordinal loss (Section IV-H) actually depends on, but the English labels are, on their face, counter-intuitive: "Moderate" ($p = 0.33$) denotes a *less* severe condition than "Mild" ($p = 0.67$), which is the reverse of everyday usage, where "mild" ordinarily implies less severity than "moderate." This inversion is inherited from the class-grouping convention in [26] rather than introduced in this paper, but it is flagged here rather than silently renamed, since relabeling would change how the results are read without being verifiable against the original per-sample annotation records.

D. Severity Formulation

Rather than defining a new severity scale, the four ELPV labels are treated as an ordinal target (Functional < Moderate < Mild < Severe, i.e., severity levels 0–3 in Table II) that increases monotonically with the original defect probability. This reformulation is the basis for the ordinal loss in Section IV-H and is evaluated against a conventional categorical treatment of the same four labels in Section VII-B. Given the class-imbalance shown in Table II — Moderate cells make up under 5% of the dataset — class-weighted loss is applied during training (Section V-C) for both the categorical and ordinal conditions, so that any observed difference between them (Section VII-B) is attributable to the ordinal formulation rather than to imbalance handling.

E. YOLO Localization Dataset

Panel/cell localization uses a separate, dedicated object-detection dataset, distinct from ELPV; ELPV does not provide bounding-box annotations and is not used to train the localizer. **[VERIFY — exact dataset identity]**: the localization experiments reported in Table XIII are consistent with a dataset of outdoor, in-the-wild solar-panel imagery of approximately 621 manually annotated images in YOLO bounding-box format, but the precise dataset name, source URL, version, and license could not be independently confirmed within this session and must be supplied by the authors before submission. This subsection is written to be completed rather than to assert an unconfirmed source.

F. YOLO Localization Dataset and Acquisition

Bounding-box annotations, where used, follow the standard YOLO representation, one line per object per image:

```
class_id x_center y_center width height
```

with all four geometric values normalized to $[0, 1]$ relative to image width and height. **[VERIFY]** items to be completed by the authors: exact image count, class list (e.g., single "panel" class versus multiple defect/panel classes), train/validation/test split, annotation tool and inter-annotator protocol, and license terms governing reuse. Panel/cell **localization** (the YOLO stage, Section IV-C) and cell-level **defect classification** (Section IV-G onward, trained on ELPV) are kept conceptually and experimentally separate throughout this paper; the localizer answers only "where," never "what defect."

G. Data Preprocessing

Cell images are channel-replicated to three channels, resized to the 224×224 input resolution used by all backbones, and normalized with standard ImageNet statistics. Rotation, horizontal flip, and colour-jitter augmentation are applied to the training split only, consistent with the preprocessing protocol reported in [27].

H. Data Splitting and Leakage Prevention

ELPV is split into approximately 70% training, 15% validation, and 15% test, stratified by class and fixed with a documented random seed prior to any model training; splitting is performed before augmentation so that augmented views of a training image cannot appear in the validation or test partitions. With 2,624 total cells, a 15% test split corresponds to approximately 394 images, which matches the test-set support total in Tables XI and XII exactly, a useful internal consistency check on the reported split. The split is image-level and stratified by class label; it is **not** confirmed to be module-disjoint. Because ELPV cells originate from only 44 modules, an image-level split cannot rule out cells from the same module appearing in both training and test partitions, which is a plausible source of optimistic bias. This study does not claim a module-disjoint split was used, and a module-disjoint protocol is identified as a stronger validation design for future work (Section X, Section XII) rather than assumed here.

I. Problem Definition

Given N candidate frames $\{x_1, \dots, x_N\}$ for a cell and a quality score $q(x) \in [0, 1]$ (Section IV-E), the selected frame and its predicted class are

$$x^* = \operatorname{argmax}_i q(x_i^*), \text{ s.t. } q(x_i^*) \geq \tau ; P(y = k | x^*) = f_{\theta}(x^*), k \in \{1, \dots, K\}$$

with $K = 4$ and an uncertainty score $u(x^*)$ obtained from repeated stochastic inference (Section IV-I). The overall pipeline is the composition $x \rightarrow x^* \rightarrow (\hat{y}, u) \rightarrow d$, where d is the maintenance-priority decision defined in Section IV-K.

TABLE III. Dataset Role Comparison

Dataset	Modality	# Images	Annotation Type	Task	Model	Role in Framework
ELPV [1], [2]	EL grayscale, 8-bit, 300×300	2,624	Continuous defect probability + module type	Cell-level defect/severity classification	Custom CNN, MobileNetV2, EfficientNet-B0, ResNet-18/50	Severity-aware defect analysis (Sections IV-G–IV-K)
YOLO localization dataset	Outdoor / in-the-wild imagery [VERIFY modality]	≈621 [VERIFY]	Bounding boxes (YOLO format)	Panel/cell localization (object detection)	YOLO [VERIFY version]	Upstream localization / region proposal (Section IV-C)

ELPV and the YOLO localization dataset are never merged; no image, label, or split is shared between them.

IV. Proposed Methodology

A. Overall Framework

The pipeline proceeds: image acquisition → panel/cell localization → candidate multi-frame acquisition → image quality assessment → quality gating → best-frame selection → preprocessing → severity-aware classification → ordinal severity prediction → uncertainty estimation → selective prediction → maintenance-priority decision (Fig. 1). Each stage is kept modular so that individual components — the classifier backbone, the uncertainty estimator — can be substituted without altering the rest of the pipeline, which also underlies the ablation protocol in Section VI.

B. Image Acquisition

The classification and severity components are trained and evaluated on ELPV (Section III-A), which provides one curated frame per cell rather than a sequence of captures. Consequently, the multi-frame experiments in Sections IV-D, VI-D, and VII-D use **synthetically constructed candidate frame pools** — each curated ELPV image is used to generate a set of N variant frames through controlled photometric and geometric perturbations standing in for repeated acquisition attempts. These are described throughout as *simulated multi-frame acquisition*, never as genuine camera bursts, drone video, or field-acquired sequences.

C. YOLO-Based Localization

A YOLO-family detector, trained on the separate localization dataset described in Section III-E/III-F, performs panel/cell localization independently of defect classification, in the same architectural role used in [5]–[8]. Its output — a bounding box — defines the region subsequently cropped for candidate-frame generation and quality scoring. This detector answers only "where is the panel/cell," never "what defect is present."

D. Quality-Aware Candidate Frame Selection

For each localized cell, N candidate frames from the simulated pool (Section IV-B) are scored with the no-reference quality function of Section IV-E; the candidate with the highest score that clears a threshold τ is selected (Section III-I). Frames below τ are treated as rejected rather than forced into classification; in the evaluation reported in Table VIII, this rejection behaviour is quantified directly (18.4% of candidate frames rejected). A cell for which no candidate clears τ within the available pool is flagged for manual inspection rather than silently misclassified.

E. Image Quality Scoring

Quality is scored with a weighted, no-reference combination of sharpness, exposure, contrast, and noise cues, in the spirit of the multi-metric gating validated for defect detection in [17]:

$$Q(x) = w_1 \cdot \text{Sharp}(x) + w_2 \cdot \text{Expo}(x) + w_3 \cdot \text{Contrast}(x) - w_4 \cdot \text{Noise}(x)$$

The weights w_1 – w_4 and threshold τ were fitted by maximizing agreement between $Q(x)$ and human usability ratings collected on a held-out set of frames, following the general calibration strategy used in [17]. The resulting quality score correlated with human ratings at Spearman $\rho = 0.87$, with 91.8% agreement, 93.1% precision, and 90.6% recall for identifying usable frames (Table VIII).

F. Preprocessing

Preprocessing of the selected frame follows the protocol already described in Section III-G (channel replication, resize to 224×224 , ImageNet normalization); no additional steps are applied at this stage.

G. Deep-Learning Classification

Five candidate backbones — a compact custom CNN, MobileNetV2, EfficientNet-B0, and ResNet-18/50 — are used as established, interchangeable feature extractors; none is presented as a contribution of this work. MobileNetV2 and EfficientNet-B0 are included primarily as lightweight options relevant to edge or on-drone deployment, while ResNet-18/50 provide direct comparability with the baselines in [26]–[28]. All backbones are initialized from ImageNet-pretrained weights and fine-tuned on the ELPV severity labels (Section III-C) with class-weighted loss.

H. Ordinal Severity Prediction

Each backbone is trained with a rank-consistent ordinal loss in the style of CORAL/CORN [15], [16],

$$L_{\text{ord}} = -\sum_n \sum_k [y_n^k \log \sigma(g_k(x_n)) + (1 - y_n^k) \log(1 - \sigma(g_k(x_n)))]$$

and compared directly, under matched backbone and training budget, against conventional categorical cross-entropy L_{CE} (RQ2, Section VII-B). The ordinal formulation targets the four severity levels of Table II as a rank-consistent sequence rather than as an unordered label set.

I. Uncertainty Estimation

MC dropout [11] with T stochastic forward passes at inference time produces a predictive distribution $\bar{p}(y|x)$ and predictive entropy

$$H(x) = -\sum_k \bar{p}(y = k|x) \log \bar{p}(y = k|x)$$

used as the model's uncertainty signal. Post-hoc temperature scaling [13] is applied to reduce over-confidence before calibration is reported; because it is a monotonic rescaling of the logits, it changes calibration without changing the predicted class ranking. A deep-ensemble estimator [12] is evaluated separately (Table IX, Section VII-F) as an alternative uncertainty mechanism; it is **not** treated as part of the main framework reported elsewhere in this paper, which uses MC dropout together with temperature scaling.

J. Selective Prediction

Predictions are ranked by $H(x)$, and the highest-entropy fraction is withheld from automatic resolution at six operating points (0%, 5%, 10%, 15%, 20%, and 25% rejected, i.e., 100% down to 75% coverage). The resulting accuracy is reported and interpreted strictly as **retained-set accuracy at a given coverage level** (Table X, Section VII-G), not as a change in the model's overall accuracy.

K. Maintenance-Priority Decision

Predicted severity $\hat{y}$ and predictive entropy $H(x)$ are combined by a threshold-based rule

$$d(x) = \pi(\hat{y}(x), H(x); \{\theta_{\text{sev}}, \theta_{\text{unc}}\}) \in \{\text{No Action, Monitor, Inspect, High Priority}\}$$

where θ_{sev} and θ_{unc} are severity and uncertainty thresholds tuned on the validation split; for example, high predicted severity combined with low uncertainty routes to "High Priority," while high severity combined with high uncertainty routes to "Inspect" for human confirmation rather than an automated high-priority alert. This mapping (evaluated in Table XV) is presented as a research-level decision-support illustration, not as an established or industry-validated maintenance protocol; it would require validation against expert inspection practice and maintenance records before operational use.

[FIGURE 2 ABOUT HERE]

Figure 2. Severity-aware ordinal classification head coupled with the MC-dropout uncertainty and maintenance-priority decision stage.

[FIGURE 3 ABOUT HERE]

Figure 3. Image quality assessment module: per-frame scoring, accept/reject gating, and best-frame selection prior to classification.

V. Experimental Setup

A. Hardware and Software Environment

[VERIFY / TBD] — exact GPU/CPU/RAM configuration and software versions are not specified in the experimental record available for this manuscript and should be inserted by the authors. The implementation is assumed to sit on a standard Python deep-learning stack (PyTorch- or TensorFlow-class framework) with OpenCV for image-quality feature extraction and scikit-learn for classical baselines, consistent with the environments described in [26], [27].

B. Dataset Split

The stratified 70/15/15 split described in Section III-H is used throughout; the test partition (≈ 394 images) is the basis for all test-set results reported in Section VII, including Tables IV, V-A, V-B, VII–XVI.

C. Training Configuration

All backbones are initialized with ImageNet-pretrained weights and fine-tuned with class-weighted loss to address the imbalance in Table II, using the AdamW optimizer and 224×224 input resolution, with rotation/flip/colour-jitter augmentation restricted to the training split. Learning rate, batch size, number of training epochs, and the number of MC-dropout inference passes T are **[VERIFY]** — not specified in the supplied experimental results and left unfilled here rather than invented.

D. Backbone Models

Custom CNN (compact baseline), MobileNetV2 and EfficientNet-B0 (edge/lightweight candidates), and ResNet-18/ResNet-50 (direct comparability with [26]–[28]) are evaluated under the identical data split and training protocol described above.

E. Evaluation Metrics

Classification is evaluated with accuracy and macro-averaged precision/recall/F1; ordinal severity with mean absolute error (MAE) and quadratic weighted kappa (QWK) [15], [16]; uncertainty and calibration with expected calibration error (ECE), Brier score, negative log-likelihood (NLL), and error-detection AUROC [11], [13]; localization with precision/recall/F1 and mAP [5]; and computational cost with parameter count, model size, and frames per second (FPS). An additional accuracy-per-million-parameters ratio is reported in Table XIV; this is a **study-specific efficiency indicator**, not a standardized benchmark metric, and is labelled as such wherever it is used.

F. Experimental Protocol

Six experimental comparisons are conducted, each isolating one design factor while holding the remainder of the pipeline fixed: (a) per-backbone baselines; (b) categorical versus ordinal loss; (c)/(d) quality gating and multi-frame candidate selection; (e) robustness under simulated degradation; (f) uncertainty-estimator comparison; (g) selective prediction; and (h) computational efficiency, together with (i) an evaluation of the maintenance-priority decision layer. These are detailed in Section VI and reported with results in Section VII.

VI. Experimental Design

Each comparison below isolates one design choice while holding the remaining pipeline fixed, so that observed differences can be attributed to a single factor rather than to confounded changes in backbone, data split, or training budget.

A. Backbone Comparison

Custom CNN, MobileNetV2, EfficientNet-B0, ResNet-18, and ResNet-50 are trained under the identical data split and training budget (Sections V-B–V-C), producing baselines directly comparable to the published ELPV figures in [1], [26]–[28].

B. Categorical vs. Ordinal Learning

EfficientNet-B0 — the strongest backbone identified in VI-A — is trained once with categorical cross-entropy and once with the ordinal loss of Section IV-H, under matched backbone, data split, and budget, isolating the effect of the ordinal formulation on severity-relevant metrics (MAE, QWK) as well as accuracy and F1 (RQ2).

C. Quality-Gating Ablation

The quality-gating threshold τ (Section IV-D) is toggled on and off while holding the candidate-generation process and classifier fixed, isolating the specific contribution of gating from the contribution of simply having multiple candidate frames.

D. Multi-Frame Selection

Four conditions are compared over the simulated candidate pools (Section IV-B): a single random frame, best-of-3 without gating, best-of-5 without gating, and quality-gated selection at $N = 3$ and $N = 5$, separating the effect of pool size from the effect of the gating function itself.

E. Robustness Evaluation

Gaussian blur, additive Gaussian noise, brightness/contrast shift, and JPEG compression are applied at five simulated severities each, following the corruption-benchmark style of [14]; the full framework and a single-frame, softmax-only baseline are compared at matched severities.

F. Uncertainty and Calibration

A softmax baseline, MC dropout, MC dropout with post-hoc temperature scaling, and a deep-ensemble estimator are compared on ECE, Brier score, NLL, and error-detection AUROC, holding the underlying classifier fixed where applicable.

G. Selective Prediction

An accuracy–coverage curve is constructed by withholding the highest-entropy predictions at six coverage levels (100% down to 75%), using the MC-dropout uncertainty scores from VI-F.

H. Computational Efficiency

Parameter count, model size, and FPS are recorded for each backbone under matched inference conditions and read alongside the accuracy figures from VI-A to characterize the accuracy–cost trade-off (RQ5).

I. Maintenance-Priority Evaluation

The four-category decision layer of Section IV-K is evaluated against severity/uncertainty-derived reference labels and reported per category and as a macro average, to characterize the decision layer’s behaviour rather than to certify it for operational use.

VII. Results

Unless stated otherwise, all figures below are computed on the held-out test split described in Section V-B ($n \approx 394$).

A. Backbone Classification Results

Table IV reports four-class accuracy, binary accuracy, and macro-F1 for the five candidate backbones. EfficientNet-B0 obtained the highest four-class accuracy (95.6%) and macro-F1 (93.5%) among the evaluated backbones, followed closely by ResNet-50 (95.1%, 93.0%) at substantially higher parameter cost (Section VII-J). The custom CNN was the weakest classifier (89.8%, 85.7%) but the least expensive. These figures are the direct counterparts of the published ELPV baselines summarized in Table I and are revisited in Section VIII-C.

TABLE IV. Per-Backbone Classification Results

Backbone	4-Class Acc.	Binary Acc.	Macro-F1	MAE	QWK
Custom CNN	89.8%	94.1%	85.7%	0.284	0.842
MobileNetV2	93.7%	96.6%	90.8%	0.214	0.901
EfficientNet-B0	95.6%	97.7%	93.5%	0.164	0.934
ResNet-18	94.4%	97.1%	92.1%	0.186	0.919
ResNet-50	95.1%	97.5%	93.0%	0.171	0.929

B. Ordinal Severity Results

Table V-A isolates the effect of the ordinal formulation on EfficientNet-B0. Replacing categorical cross-entropy with the ordinal loss raised accuracy from 94.7% to 95.6% and macro-F1 from 92.1% to 93.5%, but the larger effect is on the severity-specific metrics: MAE fell from 0.213 to 0.164 (about 23% relative reduction) and QWK rose from 0.912 to 0.934. This pattern is consistent with RQ2 and with the expectation that an ordinal objective should specifically improve rank-sensitive error, rather than only overall classification accuracy.

TABLE V-A. Categorical vs. Ordinal Severity Learning (EfficientNet-B0)

Configuration	Accuracy	Macro-F1	MAE	QWK
Cross-Entropy	94.7%	92.1%	0.213	0.912
Ordinal Loss	95.6%	93.5%	0.164	0.934

C. Component Ablation

Table V-B adds one component at a time to a single-frame, softmax-only baseline. Accuracy rises from 92.8% to 95.6% (+2.8 points) and ECE falls from 8.9% to 2.9% (roughly a two-thirds relative reduction) across the full sequence. No single step accounts for the whole improvement: the largest single accuracy increment (+1.3 points) follows the introduction of quality gating, while the largest single reduction in severity MAE (−0.041) follows the introduction of the ordinal loss.

TABLE V-B. Ablation of Proposed Components

Configuration	Accuracy	Macro-F1	MAE	QWK	ECE
Single-frame + Softmax	92.8%	89.7%	0.287	0.871	8.9%
+ Quality Gating	94.1%	91.4%	0.241	0.901	6.6%
+ Multi-frame Selection	94.6%	92.0%	0.218	0.914	5.9%
+ Ordinal Severity	95.2%	93.0%	0.177	0.928	4.7%
+ MC Dropout	95.3%	93.2%	0.173	0.930	3.8%
Full Proposed Framework	95.6%	93.5%	0.164	0.934	2.9%

[VERIFY]: the "+ MC Dropout" row reports ECE = 3.8%, which matches neither the standalone "MC Dropout" condition in Table IX (5.1%) nor the "MC Dropout + Temperature Scaling" condition (2.9%). Since the Full Proposed Framework's ECE (2.9%) matches the latter, temperature scaling is evidently included by the final row, but its separate contribution at the "+ MC Dropout" step is not cleanly attributable from the numbers as reported; this should be reconciled against the original per-step logs before submission.

D. Quality-Gated Multi-Frame Results

Table VII compares candidate-selection strategies over the simulated multi-frame pools. Increasing the number of candidate frames improves accuracy even without gating (89.9% → 93.5% → 94.0% for 1, 3, and 5 ungated candidates), but gating adds a further, separate benefit at fixed pool size: 93.5% → 94.4% at $N = 3$ and 94.0% → 95.6% at $N = 5$. This separates the contribution of simply having more candidates from the contribution of the quality function itself (RQ1).

TABLE VII. Quality-Gated Multi-Frame Acquisition

Configuration	Accepted Frames	Accuracy	Macro-F1	MAE
Random single frame	1	89.9%	85.8%	0.301
Best of 3, no quality gate	3	93.5%	90.4%	0.231
Best of 5, no quality gate	5	94.0%	91.1%	0.218

Configuration	Accepted Frames	Accuracy	Macro-F1	MAE
Quality-gated $N = 3$	3	94.4%	91.8%	0.207
Quality-gated $N = 5$	5	95.6%	93.5%	0.164

All candidate frames in this table are drawn from the synthetically constructed pools described in Section IV-B, not from genuine sequential captures. **[VERIFY]**: the "random single frame" condition (89.9%) is noticeably weaker than the "single-frame + softmax" baseline reported in Table V-B (92.8%), even though both nominally describe a single-frame condition. The most plausible explanation is that Table V-B's baseline uses the single original curated ELPV image, while Table VII's "random single frame" draws one frame at random from the synthetically degraded candidate pool; this distinction is not stated explicitly in the source results and should be confirmed by the authors.

E. Robustness Results

Table VI reports accuracy of the full framework under four simulated corruption types at severities 1, 3, and 5. Degradation is monotonic in severity for all four corruption types, and blur produces the largest overall drop (94.7% $\rightarrow$ 80.4%, a 14.3-point loss from severity 1 to 5), larger than noise (95.0% $\rightarrow$ 84.1%, 10.9 points), brightness/contrast shift (95.2% $\rightarrow$ 87.0%, 8.2 points), or JPEG compression (95.3% $\rightarrow$ 88.5%, 6.8 points). Table VI-B compares the full framework against a single-frame baseline at matched severities; the framework outperforms the baseline at every severity and corruption type, and the margin widens as corruption increases — for blur, the gap grows from +2.3 points at severity 1 to +13.6 points at severity 5. All degradations in Tables VI and VI-B are simulated; no claim is made about performance under real outdoor, weather, or drone-flight conditions.

TABLE VI. Robustness Under Simulated Degradation (Full Framework)

Corruption Type	Severity 1	Severity 3	Severity 5
Gaussian blur	94.7%	90.1%	80.4%
Additive Gaussian noise	95.0%	91.8%	84.1%
Brightness/contrast shift	95.2%	92.8%	87.0%
JPEG compression	95.3%	93.2%	88.5%

TABLE VI-B. Robustness: Baseline vs. Proposed Framework

Corruption	Severity	Baseline	Proposed	Improvement
Gaussian blur	1	92.4%	94.7%	+2.3
Gaussian blur	3	82.6%	90.1%	+7.5
Gaussian blur	5	66.8%	80.4%	+13.6
Noise	1	93.0%	95.0%	+2.0
Noise	3	85.7%	91.8%	+6.1
Noise	5	73.9%	84.1%	+10.2

Corruption	Severity	Baseline	Proposed	Improvement
Brightness/Contrast	1	93.6%	95.2%	+1.6
Brightness/Contrast	3	88.1%	92.8%	+4.7
Brightness/Contrast	5	78.4%	87.0%	+8.6
JPEG	1	94.0%	95.3%	+1.3
JPEG	3	89.7%	93.2%	+3.5
JPEG	5	81.9%	88.5%	+6.6

F. Uncertainty and Calibration

Table IX compares four uncertainty-estimation configurations. Relative to the softmax baseline (ECE 8.9%, AUROC 0.71), MC dropout improves both accuracy and calibration (95.3%, ECE 5.1%, AUROC 0.84), and adding temperature scaling improves calibration further without changing accuracy (ECE 2.9%, AUROC 0.88), consistent with temperature scaling being a monotonic rescaling of the logits. A deep ensemble achieves marginally better figures still (95.8% accuracy, ECE 2.5%, AUROC 0.90) but is reported as an **alternative** configuration rather than a component of the main framework, which uses MC dropout with temperature scaling (Section IV-I); the ensemble's requirement to train and run multiple independent networks is the practical reason it is not adopted as the default.

TABLE IX. Uncertainty Estimation and Calibration

Configuration	Accuracy	ECE	Brier	NLL	Error-Detection AUROC
Softmax baseline	92.8%	8.9%	0.142	0.318	0.71
MC Dropout	95.3%	5.1%	0.109	0.241	0.84
MC Dropout + Temperature Scaling	95.3%	2.9%	0.096	0.198	0.88
Deep Ensemble (alternative)	95.8%	2.5%	0.091	0.187	0.90

G. Selective Prediction

Table X reports retained-set accuracy as increasing fractions of the highest-entropy predictions are withheld. Accuracy on the retained subset rises monotonically from 95.3% at 100% coverage (no rejection) to 99.3% at 75% coverage (25% rejected). These are **retained-set accuracies conditioned on rejection**, not the model's overall accuracy; the trade-off is that higher rejection improves confidence in the automatically resolved cases at the cost of routing a larger share of cells to manual review (RQ3).

TABLE X. Accuracy vs. Selective Prediction (Coverage)

Coverage	Rejected	Retained-Set Accuracy
100%	0%	95.3%
95%	5%	96.8%
90%	10%	97.7%

Coverage	Rejected	Retained-Set Accuracy
85%	15%	98.4%
80%	20%	99.0%
75%	25%	99.3%

H. Class-Wise Severity Results and Confusion-Matrix Audit

Table XI reports the originally supplied class-wise precision, recall, and F1, and Table XII reports the corresponding confusion matrix, exactly as supplied and without modification. **[VERIFY — metric consistency audit]**: the confusion matrix in Table XII sums to 394 samples with 378 correct predictions on the diagonal ($220 + 15 + 40 + 103$), implying an overall accuracy of 95.94% rather than the 95.6% reported in Tables IV, V-A, V-B, VII, and XVI. Recomputing precision, recall, and F1 directly from Table XII's row and column sums gives values that diverge from Table XI by up to several points for the minority classes — for example, a matrix-derived Moderate-class precision of 75.0% (15 of 20 predicted-Moderate cells correct) against the reported 84.2%, and a matrix-derived macro-F1 of about 90.5% against the reported 91.5%. Because Table XI and Table XII cannot both be exactly correct simultaneously, neither is silently altered here. For the remainder of this paper, the 95.6% four-class accuracy is retained as the headline figure because it is independently corroborated across five other tables (IV, V-A, V-B, VII, XVI), while Tables XI and XII, and the class-wise figures derived from them, should be reconciled against the original per-sample prediction log before camera-ready submission.

TABLE XI. Class-Wise Severity Performance (as reported) [VERIFY]

Class	Precision	Recall	F1	Support
Functional	97.4%	98.0%	97.7%	225
Moderate	84.2%	78.9%	81.5%	18
Mild	91.6%	89.7%	90.6%	45
Severe	96.0%	96.7%	96.3%	106
Macro Average	92.3%	90.8%	91.5%	394

TABLE XII. Confusion Matrix (Test Set, n = 394) [VERIFY]

Actual \ Predicted	Functional	Moderate	Mild	Severe
Functional	220	3	2	0
Moderate	1	15	2	0
Mild	1	2	40	2
Severe	0	0	3	103

I. YOLO Localization Results

Table XIII reports the localization detector's performance on its own held-out split. The detector reached 96.1% mAP@0.50 and 84.7% mAP@0.50:0.95, with 94.2% precision, 93.1% recall, and a 4.3% localization failure rate at 32 FPS. **[VERIFY]**: Table XIII's own heading refers to "panel/cell localization" without disambiguating which level (panel or individual cell) the reported metrics describe; this paper uses the two terms deliberately throughout (Sections III-E–III-F, IV-C) and flags the ambiguity here rather than assuming one reading. This detector's 32 FPS throughput is markedly lower than any of the classification backbones in Table XIV, consistent with it operating at a different input resolution and stage of the pipeline.

TABLE XIII. Panel/Cell Localization

Metric	Value
Precision	94.2%
Recall	93.1%
F1	93.6%
mAP@0.50	96.1%
mAP@0.50:0.95	84.7%
Inference FPS	32
Localization failure rate	4.3%

J. Computational Efficiency

Table XIV lists parameters, model size, accuracy, FPS, and an accuracy-per-million-parameters ratio for each backbone. The ratio is computed as four-class accuracy (Table IV) divided by parameter count in millions and is internally consistent with the reported values (e.g., EfficientNet-B0: $95.6/5.3 = 18.04$); it is a study-specific efficiency indicator introduced for this comparison, not a standardized benchmark metric. The custom CNN has the highest such ratio (42.76) and the highest FPS (92) but the lowest accuracy (89.8%); EfficientNet-B0 offers the best accuracy among all backbones (95.6%) while remaining substantially cheaper than ResNet-50 (5.3M vs. 25.6M parameters, 68 vs. 38 FPS) for a 0.5-point accuracy advantage (RQ5).

TABLE XIV. Computational Efficiency

Backbone	Parameters	Model Size	Accuracy	FPS	Accuracy / M Params
Custom CNN	2.1M	8.4 MB	89.8%	92	42.76
MobileNetV2	3.5M	14.0 MB	93.7%	80	26.77
EfficientNet-B0	5.3M	21.2 MB	95.6%	68	18.04
ResNet-18	11.7M	46.8 MB	94.4%	58	8.07
ResNet-50	25.6M	102.4 MB	95.1%	38	3.71

K. Maintenance-Priority Results

Table XV reports precision, recall, and F1 for the four-category maintenance-priority decision (Section IV-K). "No Action" is identified most reliably (F1 98.1%), while "Inspect" — the category intended to capture

uncertain, moderate-risk cases — is identified least reliably (F1 85.5%), which is consistent with this category being defined precisely for the ambiguous middle ground between clear cases. The macro-averaged F1 (91.1%) is internally consistent with the four per-category values.

TABLE XV. Maintenance-Priority Decision Performance

Category	Precision	Recall	F1	Interpretation
No Action	97.8%	98.5%	98.1%	Functional / negligible defect
Monitor	89.6%	87.2%	88.4%	Low-risk degradation
Inspect	86.4%	84.7%	85.5%	Uncertain / moderate-risk
High Priority	93.2%	91.5%	92.3%	Severe with reliable prediction
Macro Average	91.8%	90.5%	91.1%	—

This mapping is a research-level decision-support illustration (Section IV-K); it is not presented as an established industry maintenance standard.

L. Overall Results

Table XVI collects the headline figures already reported in Tables IV–XV. It is presented as a summary index rather than a new measurement, and it inherits the verification flags already raised for its component tables (notably Section VII-C and VII-H).

TABLE XVI. Overall Performance Summary

Metric	Value
4-class accuracy	95.6%
Binary accuracy	97.7%
Macro-F1	93.5%
Severity MAE	0.164
QWK	0.934
ECE	2.9%
Error-Detection AUROC	0.88
Accuracy @ 80% coverage	99.0%
Quality-gating Spearman ρ	0.87
Localization mAP@0.50	96.1%
Blur severity-5 accuracy	80.4%
EfficientNet-B0 throughput	68 FPS

[FIGURE 4 ABOUT HERE]

Figure 4. Confusion matrix heatmap (Table XII) and reliability diagram for the full proposed framework. (*To be plotted by the authors directly from Tables IX and XII; no image is generated in this manuscript.*)

VIII. Discussion

A. Main Findings

Across the completed experiments, four patterns stand out: (1) EfficientNet-B0 with the ordinal loss was the strongest configuration on every severity-relevant metric; (2) each proposed component — quality gating, multi-frame selection, ordinal severity, and MC-dropout calibration — contributed a distinct, non-overlapping share of the total gain shown in Table V-B; (3) uncertainty estimation and selective prediction behaved as intended, improving retained-set accuracy monotonically as coverage decreased; and (4) the framework's advantage over a single-frame baseline widened, rather than narrowed, as simulated image degradation increased.

B. Effect of Quality Gating (RQ1)

Quality gating improved accuracy beyond what additional candidate frames alone provided (Section VII-D): at a fixed pool of five candidates, gating raised accuracy from 94.0% to 95.6%. The gating function's quality score also correlated with human usability judgments at Spearman $\rho = 0.87$ (Table VIII), which supports treating it as a meaningful, if imperfect, proxy for frame usability rather than an arbitrary filter. These results should be read as evidence that acquisition-quality awareness helps under the *simulated* multi-frame protocol used here; they do not by themselves demonstrate benefit under genuine field acquisition, which was not tested.

C. Effect of Ordinal Learning (RQ2)

The ordinal loss reduced MAE by roughly 23% relative to categorical cross-entropy ($0.213 \rightarrow 0.164$) and raised QWK from 0.912 to 0.934, with smaller gains in accuracy and F1 (Table V-A). Because both conditions used the same class-weighted loss to address the imbalance in Table II, the MAE/QWK improvement is attributable to the ordinal formulation itself rather than to imbalance handling, though this holds only for the single backbone (EfficientNet-B0) on which the comparison was run.

D. Effect of Uncertainty Estimation (RQ3)

MC dropout combined with temperature scaling reduced ECE from 8.9% (softmax baseline) to 2.9% and raised error-detection AUROC from 0.71 to 0.88 (Table IX). An AUROC of 0.88, not 1.0, means the uncertainty signal is informative but not a perfect predictor of error; this is reflected in the selective-prediction curve (Table X), where even at 25% rejection the retained-set accuracy is 99.3%, not 100%. The practical reading is that uncertainty-based rejection can usefully route a subset of cases to human review, not that it eliminates errors.

E. Robustness (RQ4)

Blur produced the largest simulated degradation of the four corruption types tested, and the proposed framework's margin over the single-frame baseline grew with corruption severity for every corruption type (Table VI-B). This is consistent with the framework's quality-gating and multi-frame components being more useful precisely when individual frames are more likely to be compromised; it does not establish which backbone is most robust in general, since only the full framework and one baseline were compared under corruption, and the results describe simulated corruption rather than real optical or environmental degradation.

F. Accuracy–Efficiency Trade-Off (RQ5)

EfficientNet-B0 provided the best four-class accuracy among the tested backbones while requiring under half the parameters and roughly twice the throughput of ResNet-50 for a 0.5-point accuracy advantage (Table XIV). This makes it, in this comparison, a favourable balance of accuracy and cost rather than a universally optimal choice; the custom CNN remains preferable if the priority is minimizing parameters or maximizing throughput at the cost of roughly six accuracy points.

G. Practical Interpretation

The maintenance-priority decision layer (Table XV) is weakest precisely on the "Inspect" category that is meant to capture ambiguous cases, which is a reasonable property for a decision-support tool but means its output should not be treated as a final maintenance determination without human review, especially for cases routed to "Inspect." Combined with the selective-prediction results, a plausible operational reading is that the system is best used to triage clearly resolved cases (high confidence, "No Action" or "High Priority") automatically, while routing uncertain or borderline cases to inspection, rather than to fully automate maintenance decisions.

IX. Explainability Analysis

A Grad-CAM-based qualitative analysis was planned to visualize which image regions drive the ordinal severity head's predictions, applied to the final convolutional block of the selected backbone. No quantitative explainability metric is reported in this study, and no saliency maps or region-overlap statistics are included in the supplied experimental record; consequently, this section is left as a description of planned qualitative analysis rather than a set of results, and no heatmaps or explainability figures are fabricated here. If a qualitative Grad-CAM analysis is later performed, it should assess, at minimum, whether attention concentrates on defect regions (cracks, disconnected fingers) for confidently correct predictions and whether high-entropy predictions (Section IV-I) coincide with visually ambiguous regions, but both remain future work rather than reported findings.

X. Limitations

ELPV contains only 2,624 cell images drawn from 44 modules, and the Moderate class contains just 123 samples (Table II); conclusions about that class, and about the ordinal-vs-categorical comparison in general, should be read with this small-sample caveat in mind. ELPV is single-modality, grayscale EL imagery; none of the reported results should be extrapolated to RGB or thermal inspection without separate validation. ELPV does not provide genuine multi-frame acquisition sequences, so the multi-frame and quality-gating experiments (Sections IV-B, VII-D) use synthetically constructed candidate pools, and the robustness experiments (Section VII-E) use simulated corruption; neither is a substitute for field- or drone-acquired data under real optical and environmental variation. The data split (Section III-H) is image-level and stratified, not confirmed module-disjoint, so correlated cells from the same module may appear across training and test partitions, which is a plausible source of optimistic bias not ruled out here. The YOLO localization dataset is a separate visual domain from ELPV (Section III-E), and this study does not evaluate whether the classifier trained on ELPV generalizes to cells localized from that domain end-to-end. No field deployment, drone flight validation, or live inspection trial was conducted; all numbers in Section VII come from held-out offline evaluation. The maintenance-priority mapping (Section IV-K, Table XV) is an illustrative decision-support layer that has not been validated against real maintenance outcomes, technician workflows, or safety standards. Finally, this manuscript also carries several unresolved verification items — a confusion-matrix/class-wise metric mismatch (Section VII-H), an ECE inconsistency within the ablation sequence (Section VII-C), an

unconfirmed exact identity for the YOLO dataset (Section III-E), and several unspecified training hyperparameters (Section V-C) — which are flagged rather than silently resolved and should be closed out from the original experiment logs before submission.

XI. Conclusion and Future Work

This paper evaluated an integrated, severity-aware and uncertainty-guided framework for automated PV cell inspection that couples quality-gated multi-frame acquisition with an ordinal severity classifier, MC-dropout uncertainty estimation, and a maintenance-priority decision layer. On the ELPV benchmark, EfficientNet-B0 with an ordinal loss achieved the strongest reported configuration (95.6% four-class accuracy, 93.5% macro-F1, MAE 0.164, QWK 0.934), improving on an otherwise identical categorical classifier primarily on the severity-specific metrics MAE and QWK. An incremental ablation attributed measurable, distinct contributions to quality gating, multi-frame selection, ordinal severity learning, and MC-dropout calibration, jointly lowering ECE from 8.9% to 2.9% and raising error-detection AUROC from 0.71 to 0.88. Selective prediction showed that withholding the 20% most uncertain cases raised retained-set accuracy to 99.0%, and the framework degraded less than a single-frame baseline under every tested simulated corruption, with the largest advantage under the strongest degradation. A separate YOLO-based localizer reached 96.1% mAP@0.50 on its own dataset. These results were obtained on a single, comparatively small, laboratory-curated benchmark using simulated multi-frame and degradation protocols, with an unresolved confusion-matrix/class-wise metric inconsistency flagged rather than corrected, and they should be read within those limits (Section X).

Future work includes replacing the synthetic multi-frame and degradation protocols with genuine field- or drone-acquired multi-frame bursts; a module-disjoint train/test split to rule out the source-module leakage risk identified in Section III-H; evaluation on larger and more diverse EL datasets; multimodal fusion of RGB, thermal, and EL imagery; end-to-end evaluation of the YOLO localizer feeding the ELPV-trained classifier; edge deployment of the more compact backbones for on-drone inference; a more exhaustive comparison of MC dropout, deep ensembles, and evidential deep learning for uncertainty estimation under distribution shift; and expert-validated refinement of the maintenance-priority mapping against real maintenance records.

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